

Data Analysis - GIS

Lab 2

Background:

We will be performing some basic GIS operations in this assignment using Vector and Raster files. Vector and Raster data are the two most common types of spatial data. You can relate some parts of this assignment to the Lecture 2. As mentioned in Lecture 2, raster files can be generated in multiple ways such as; satellite picture, scanned map, etc. In this assignment, scanned old maps are used as raster files. Vector data related to demographics will be used in this assignment.

Purpose:

The purpose of this assignment is divided into two tasks. First task is to perform georeferencing operations with a raster file and create print layout. A raster file may not contain GIS information. Therefore, using georeferencing, you can assign GIS information to a raster file and align it with a vector file. Second task is to perform proximity analysis using school location and demographic data to analyse child population's numbers in proximity of each of the school.

Task1:

1. Georeference Sportfält1990.tiff file with the layer **motor_vagar**.
2. Create point and polygon features representing buildings, trees, lakes, etc.
3. Add a text field to the newly created features.
4. Generate label using the text field of these features.
5. Create a print layout following the cartographic rules (i.e. add legend, title and scalebar).
6. **Export the map as a PDF** and name it as Task1_A2.

Task2:

1. Select one kommun from shapefile **Kommuner_Sverige**. For example, Borlänge kommun. Create new shapefile using this selection.
2. Select demographics data of the selected kommun from the shapefile **Bef_Alder_region** and create new shapefile only for demographics in the selected kommun.
3. Identify location of elementary schools, "grundskolor" for the selected kommun. Use the official kommun websites to find this information. Look for X and Y coordinates for location of each of the school.
4. Create a new point layer showing the location of the elementary schools for the

selected municipality.

5. Use symbology and create a graduated map using Ald7_15 attribute column.
6. Using the newly created point layer of schools (i.e. from Step 5), create buffers with 1000m and 1500m. You will use these buffers to analyse number of children being resided in proximity to each of these schools. Save buffered regions as two separate layers.
7. Create map(s) showing number of children in 1000m and 1500m proximity of elementary schools. (Here, you need to make trade-off between amount of the information you have and readability of the map). **Two maps should be submitted in PDF format** (refer to Exercise 2-Instructions for help). Name the maps as Task2_Map1_A2 and Task2_Map2_A2.

Task3 (Using Python in Qgis):

- 1- With Using Python add the Road_Centerlines.shp and change the color to the black and change the name to road.
- 2- With Using Python add the Building.shp and change the color to the orange and change the name to Building.
- 3- With Using Python add the SIDEWALKS.shp and change the color to the red and change the name to sidewalks.
- 4- With Using Python add the INTERSECTIONS.shp and change the color to the gray and change the name to Intersections.
- 5- With Using Python Export Intersections point file into Intersections.csv file and save the file.

The Python code file needs to be submitted.

Material:

All material for the tasks can be found in lab2.zip in Canvas.

Deadline:

Assignment needs to be submitted by 17/5 by 23:59 in Canvas.

Grade:

This assignment is graded as U/G.

Note: In this task you work individually.